

Total Questions: 35

LEVEL – II

Time: 1.5 hr.
Total Marks: 70

SECTION - 1

- This section contains **FIFTEEN (15)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 2

- This section contains **EIGHT (08)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each correct answer you will get **TWO** marks, if all the options are correct. If the correct answers chosen are less than the actual correct answers, partial marking (basis on weightage) will be given.
- There are **NO NEGATIVE** marks for wrong answers.

SECTION - 3

- This section contains **NINE (09)** questions of **PASSAGE TYPE OR COMPREHENSION TYPE**.
- Under each paragraph there will be three questions that has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers.

SECTION - 4

- This section contains **THREE (03)** questions. These questions are **MATCHING TYPE BASED QUESTIONS**.
- For each correct answer you will get **TWO** marks. There are **NO NEGATIVE** marks for wrong answers

Section - 1

1. A student looks at a standard analog clock reflected in a plane mirror. The reflection shows the time as 9:25. What is the actual time shown on the clock face?
A) 3:35 B) 3:25
C) 2:25 D) 2:35
2. A device is labeled to have a power of 2 kilowatts (kW). If this device is connected to a mains supply and consumes 7200 joules of energy, how long was the device switched on?

- A) 3.6 seconds B) 3600 seconds
C) 0.27 seconds D) 1 hour
3. Which of the following options correctly identifies the true (T) and false (F) statements among I-IV?
I. During a physical change, new substances are formed.
II. Physical changes are always permanent and cannot be reversed.
III. The weight of a substance changes during a physical change.

IV. Ice changing into water does not involve formation of a new substance.

	I	II	III	IV
A)	T	F	F	T
B)	F	T	F	T
C)	F	F	T	T
D)	F	F	F	T

4. A student performs two experiments:

- Experiment I: A metal reacts with dilute acid and produces a gas that burns with a 'pop' sound.
- Experiment II: A white solid reacts with dilute acid and produces a gas that turns lime water milky.

Which combination correctly identifies the type of reactions involved?

- A) I: Metal + Acid $\rightarrow$ Salt + H₂; II: Carbonate + Acid $\rightarrow$ Salt + H₂O + CO₂
- B) I: Metal oxide + Acid $\rightarrow$ Salt + H₂O; II: Metal + Acid $\rightarrow$ Salt + H₂ + CO₂
- C) I: Carbonate + Acid $\rightarrow$ Salt + CO₂; II: Metal oxide + Acid $\rightarrow$ Salt + H₂O
- D) I: Metal + Acid $\rightarrow$ Salt + CO₂; II: Carbonate + Acid $\rightarrow$ Salt + H₂

5. A student sets up an experiment where a potted plant is kept in a dark room for 48 hours to de-starch it. Then, one leaf is partially covered with a strip of black paper, and the plant is moved to bright sunlight for 8 hours. After boiling the leaf in alcohol and adding iodine solution, what will the student observe, and why?

- A) The entire leaf turns blue-black because the plant had already stored starch in its stem.
- B) Only the covered part turns blue-black because starch moves from the roots to the covered area.

C) The exposed parts turn blue-black, while the covered part stays pale, proving light is essential for starch synthesis.

D) The entire leaf remains pale because boiling in alcohol destroys the starch molecules completely.

6. A person consumes a piece of plain boiled rice and chews it for a long time without swallowing. After a minute, they notice the rice starts tasting slightly sweet. Which of the following best explains this chemical change in the mouth?

- A) The physical grinding of the teeth turns the complex starch into salt.
- B) Salivary amylase breaks down the complex starch into simpler sugars like maltose.
- C) The acid from the stomach comes up to the mouth to digest the rice.
- D) The rice reacts with the oxygen in the air to produce glucose crystals on the tongue.

7. A child gets a deep scratch while playing. Initially, it bleeds, but after two minutes, a dark red "clot" form. If the child's blood lacked Fibrinogen (a protein) and Platelets, what would be the physiological result?

- A) The blood would change color from red to blue.
- B) The bleeding would not stop easily, leading to a dangerous loss of blood volume.
- C) The scratch would heal instantly because platelets usually slow down healing.
- D) The heart would start pumping blood in the opposite direction to save the blood.

8. A metal block is heated and then allowed to slide on a rough surface until it comes to rest. Which of the following statements are correct?

A) Heat energy decreases the internal energy of the block
 B) Work done against friction converts heat into mechanical energy
 C) The temperature of the block must always decrease while sliding
 D) Energy is conserved during the entire process

9. Ravi tries to pass electric current through two samples given below. He observes that only one of them allows electricity to pass. Why does this difference occur?



A) Pure water contains more energy
 B) Salt water contains dissolved salts that conduct electricity
 C) Distilled water has higher boiling point
 D) Salt water freezes faster

10. Earthworms and Frogs both use their skin for respiration. However, for this to be effective, their skin must be kept constantly moist. What is the logical reason for this requirement?

A) Dry skin is too hard for the animal to move.
 B) Gases (Oxygen and Carbon Dioxide) can only diffuse across the respiratory surface if they are first dissolved in a thin layer of moisture.

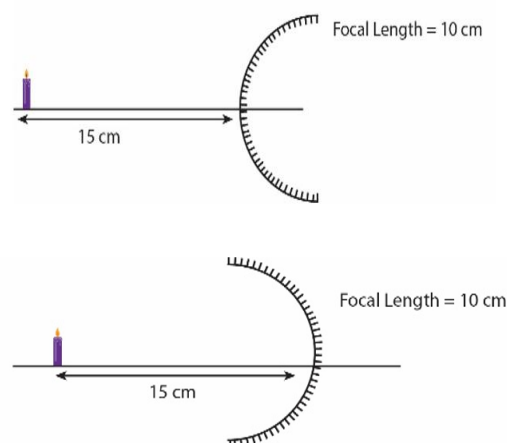
C) Moisture acts as a shield to reflect sunlight so the animal doesn't get a sunburn.

(D) The water on the skin is used to produce sugar for the animal's muscles.

11. In an experiment, a Spirogyra (green alga) filament is cut into 5 equal pieces. In another tank, a Rose flower is cut into 5 pieces. What will be the observation after 10 days?

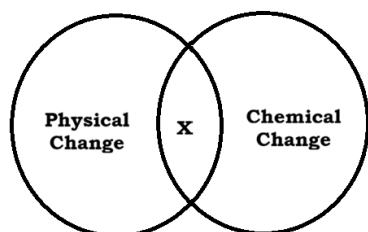
A) All 5 pieces of Spirogyra grow into new filaments, but the 5 pieces of the rose flower decay.
 B) The rose flower pieces grow into 5 new rose bushes, but the Spirogyra dies.
 C) Both grow into new organisms because all plant parts are the same.
 D) Neither grows because reproduction can only happen through seeds.

12. Which statement correctly describes the images?



A) The concave mirror image is virtual and diminished; the convex mirror image is real and magnified.
 B) Both images are real and inverted.
 C) The concave mirror image is real and magnified; the convex mirror image is virtual and diminished.
 D) Both images are virtual and erect.

13. Observe the below Venn diagram. Which of the following can be placed in the X region?



- A) Formation of new substance
 B) Change in state
 C) Energy change
 D) Irreversibility
14. A student removes the "Anthers" from a bisexual flower before they mature, but leaves the "Pistil" intact. She then covers the flower with a bag. Later, she dusts pollen from a different plant onto the stigma of this flower. What is this process called, and what is its goal?
- A) Self-pollination; to ensure the plant is a clone.
 B) Artificial hybridization (Cross-pollination); to combine desirable traits from two different parent plants.
 C) Fragmentation; to break the flower into new plants.
 D) Germination; to make the flower grow into a leaf.
15. The "Alveoli" in the lungs are thin-walled, moist, and surrounded by a vast network of capillaries. If these alveoli were replaced by one large, smooth-walled sac, how would the rate of diffusion of gases change?
- A) The rate would increase because there are fewer walls to cross.

- B) The rate would decrease significantly because the total surface area for gas exchange would be much smaller.
 C) The rate would stay the same as long as the person keeps breathing.
 D) The rate would decrease because a large sac would be too heavy for the diaphragm to move.

Section - 2

16. Two resistors 6Ω and 3Ω are connected in series to a 9 V battery. Which are correct?
- A) Total resistance = 9Ω
 B) Current = 1 A
 C) Voltage across 6Ω = 6 V
 D) Voltage across 3Ω = 3 V
17. A student tested four samples P, Q, R, S using methyl orange indicator. Which of the following observations are correct?



- A) P shows red colour
 B) Q shows red colour
 C) R shows yellow colour
 D) S shows yellow colour
18. Plants require various minerals for healthy growth. Which of the following correctly describes the role of specific nutrients?
- A) Nitrogen is essential for the synthesis of proteins and chlorophyll.
 B) Phosphorus is a key component of ATP and helps in root development.

- C) Magnesium is the central element found in the chlorophyll molecule.
- D) Plants can absorb atmospheric nitrogen directly through their stomata.

19. In a circuit, an ammeter and voltmeter are interchanged mistakenly.

- A) Ammeter (now parallel) may get damaged
- B) Voltmeter (now series) reduces current
- C) Circuit still works normally
- D) Voltmeter shows wrong reading

20. How many of the below matches are correct?

- (A)   Chemical Change
- (B)   Chemical Change
- (C)   Chemical Change
- (D)   Physical Change

21. In the small intestine, digestion is completed and absorption begins. Which of the following are adaptations of the small intestine for efficient absorption?

- A) It is very long (about 7.5 meters) to provide more time for absorption.

- B) Its inner walls have villi that significantly increase the surface area.
- C) Villi are richly supplied with blood capillaries to transport absorbed food.
- D) It has a thick layer of mucus to prevent the food from touching the walls.

22. Consider a heart with a heart rate of 75 beats per minute and a stroke volume of 70 mL.

Which of the following statements are mathematically or biologically correct?

- A) The cardiac output is 5,250 mL (5.25 Liters) per minute.
- B) Each heartbeat (cardiac cycle) takes exactly 0.8 seconds.
- C) If the heart rate increases, the cardiac output will also increase, provided the stroke volume stays the same.
- D) Pulse rate is always twice the heart rate.

23. Inhalation and exhalation are the two phases of breathing. Which of the following occur during the process of EXHALATION?

- A) The diaphragm moves upwards and relaxes into its dome shape.
- B) The ribcage moves downwards and inwards.
- C) The volume of the chest cavity decreases, increasing the air pressure inside.
- D) Air rushes into the lungs because the pressure inside is lower than the outside.

Section - 3

COMPREHENSION-1

Most of the water on Earth (i) _____ and is not suitable for drinking. A large portion of fresh water is (ii) _____ and hence unavailable for use. Only a very small amount of water, about (iii) _____, is

available in rivers, lakes and groundwater. Rivers are important because they help in (iv) _____ for human use.

24. Which of the following correctly fills blank (i)?

- A) is fresh B) is salty
C) is frozen D) is potable

25. Which combination correctly fills blanks (iii) and (iv)?

- A) 10%, irrigation
B) 0.01%, storing water in dams
C) 1%, desalination
D) 97.4%, evaporation

26. A group of students visits a mountainous region shown below



observes that water present cannot be used for drinking or agriculture.

Which option correctly explains this situation using blank (ii)?

- A) Fresh water is evaporated into the air
B) Fresh water is frozen in glaciers
C) Fresh water is mixed with salt
D) Fresh water is stored in rivers

COMPREHENSION-2

Ravi conducted two different experiments to understand transportation in plants and animals.

In the first experiment, he took a healthy potted plant and removed a ring of bark from its stem (girdling experiment). After a few days, he observed swelling above the cut region while the lower part of the plant started weakening.

In another setup, Ravi placed a plant with its roots dipped in colored water. After a few hours, he noticed that the veins of the leaves turned the same color as the water.

Meanwhile, Ravi also studied human blood circulation using a model. He observed that blood flows through different types of vessels and carries oxygen and nutrients to cells while removing wastes. He noted that when a person has low hemoglobin levels, they often feel tired and weak.

27. The swelling observed above the cut region in the girdling experiment is due to:

- A) Accumulation of water
B) Accumulation of food
C) Increased transpiration
D) Increased mineral absorption

28. The experiment with colored water demonstrates:

- A) Bidirectional transport in phloem
B) Upward movement of water in xylem
C) Role of stomata in gas exchange
D) Storage of food in roots

29. Which blood component helps the body fight infections?

- A) Red blood cells B) Platelets
C) White blood cells D) Plasma

COMPREHENSION-3

A delivery drone first moves straight for 60 m, then turns and follows a curved path of 80 m, and finally returns directly to its starting point in 100 m. The entire motion takes 40 seconds, with varying speed during turns.

30. The average speed and magnitude of average velocity of the drone are respectively:

- A) 6 m/s, 0 m/s B) 6 m/s, 2.5 m/s
C) 5 m/s, 0 m/s D) 4 m/s, 2 m/s

31. If the drone maintains constant speed in magnitude but continuously changes direction during the curved path, which statement is correct?

- A) Acceleration is zero because speed is constant
 B) Acceleration is non-zero due to change in direction
 C) Velocity is constant
 D) Both acceleration and velocity are zero

32. Assertion (A): The average velocity of the drone is zero.

Reason (R): The drone returns to its starting point.

- A) Both A and R are true, and R is correct explanation
 B) Both A and R are true, but R is not correct explanation
 C) A is true, R is false
 D) A is false, R is true

Section - 4

33.

COLUMN A	COLUMN B
A) Pollination	P) Fusion of gametes
B) Fertilization	Q) Transfer of pollen
C) Seed formation	R) Development after fertilization
D) Germination	S) Growth of seed into plant

- A) A-Q, b-P, c-R, d-S
 B) A-P, b-Q, c-S, d-R
 C) A-Q, b-R, c-P, d-S
 D) A-R, b-P, c-Q, d-S

34.

COLUMN -A	COLUMN-B
A) Ammeter	P) High resistance device connected in parallel.
B) Voltmeter	Q) Contains a zinc plate that can be moved up and down.
C) Bichromate Cell	R) Low resistance device connected in series.
D) Dry Cell	S) Uses a chemical paste of Ammonium Chloride.

- A) A-R, B-P, C-Q, D-S
 B) A-P, B-R, C-S, D-Q
 C) A-R, B-P, C-S, D-Q
 D) A-Q, B-P, C-R, D-S

35.

COLUMN -A	COLUMN-B
A) Muriatic acid	P) Dibasic base
B) Oil of vitriol	Q) Tribasic acid
C) Orthophosphoric acid	R) Monobasic acid
D) Slaked lime	S) Dibasic acid

- A) A - P, B - Q, C - R, D - S
 B) A - Q, B - P, C - R, D - S
 C) A - R, B - S, C - Q, D - P
 D) A - Q, B - S, C - R, D - P